6	(a)	The magnetic force acting on the electrons provides its centripetal force. $F_c = ma_c$ $Bqv = \frac{mv^2}{r}$ $8.0 \times 10^{-4} \times 1.60 \times 10^{-19} = \frac{9.11 \times 10^{-31} \times v}{6.4 \times 10^{-2}}$ $v = 8.992 \times 10^6 \approx 9.0 \times 10^6 \text{ ms}^{-1}$	C1 A1
	(b)(i)	For the electron to pass through undeviated, the net force acting on it must be zero. This requires the electric force and the magnetic force to be equal in magnitude but opposite in direction. Since the magnetic force on the electron is directed downwards, the electric force must act upwards to balance it. Since an electric force acting on a negatively charged particle is in the opposite direction to that of the electric field lines, the electric field must be directed downwards.	A1
	(b)(ii)	$F_E = F_B$ $qE = Bqv$ $q \times E = 8.0 \times 10^{-4} \times q \times 8.992 \times 10^6$ $E = 7.193 \times 10^3 \approx 7.2 \times 10^3 \text{ NC}^{-1}$	A1
	(c)	The magnetic force acting on the positron is now directed upwards while the electric force acting on the positron is now directed downwards. The resultant force acting on the positron remains unchanged. Hence the positron continues to move in a straight line or undeviated.	B1 B1